1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = 2*(L+W) and Area = L*W.

```
D6_96715_bhagyesh>CREATE PROCEDURE rectangle_calc()
    -> BEGIN
    ->     DECLARE L DOUBLE DEFAULT 10;
    ->     DECLARE W DOUBLE DEFAULT 5;
    ->     DECLARE perimeter DOUBLE;
    ->     DECLARE area DOUBLE;
    ->
    ->     SET perimeter = 2 * (L + W);
    ->     SET area = L * W;
    ->
    ->     SELECT L AS Length,
    ->            W AS Width,
    ->            perimeter AS Perimeter,
    ->            area AS Area;
    -> END $$
Query OK, 0 rows affected (0.02 sec)

D6_96715_bhagyesh>
D6_96715_bhagyesh>DELIMITER ;
D6_96715_bhagyesh>
D6_96715_bhagyesh>CALL rectangle_calc(2,3);
ERROR 1318 (42000): Incorrect number of arguments fo
le_calc; expected 0, got 2
D6_96715_bhagyesh>CALL rectangle_calc();
+--------+-------+-----------+------+
| Length | Width | Perimeter | Area |
+--------+-------+-----------+------+
|     10 |     5 |        30 |   50 |
+--------+-------+-----------+------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
D6_96715_bhagyesh>CREATE PROCEDURE SP_CAL(IN p_n1 INT,  OUT p_SUQ INT, OUT p_CUBE
 INT)
    -> BEGIN
    ->  SET p_SUQ =p_n1*p_n1;
    ->  SET p_CUBE =p_n1*p_n1*p_n1;
    ->  INSERT INTO temp VALUES(p_SUQ,p_CUBE);
    ->  END $$
Query OK, 0 rows affected (0.03 sec)

D6_96715_bhagyesh>DELIMITER ;
D6_96715_bhagyesh>call SP_CAL(2,@p_SUQ,@P_CUBE);
Query OK, 1 row affected (0.03 sec)

D6_96715_bhagyesh>SELECT * FROM temp;
+------+------+
| S    | C    |
+------+------+
|    4 |    8 |
+------+------+
1 row in set (0.00 sec)
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are

C= (F-32) *5/9 F= 9/5*C + 32

```
D6_96715_bhagyesh>CREATE PROCEDURE SP_CAL(
    -> IN f INT, IN c INT, OUT FER INT, OUT TEM INT )
    -> BEGIN
    ->
    -> SET FER = 9/5 * c + 32;
    -> SET TEM = (f - 32) * 5/9;
    ->
    -> INSERT INTO TEMPERATURE(F,T)
    -> VALUES(FER,TEM);
    ->
    -> END $$
Query OK, 0 rows affected (0.01 sec)

D6_96715_bhagyesh>
D6_96715_bhagyesh>DELIMITER ;
D6_96715_bhagyesh>CALL TEMPERATURE(4,3,@FER,@TEM);
ERROR 1305 (42000): PROCEDURE classwork.TEMPERATURE does not exist
D6_96715_bhagyesh>CALL SP_CAL(4,3,@FER,@TEM);
Query OK, 1 row affected (0.02 sec)

D6_96715_bhagyesh>SELECT * FROM SP_CAL;
ERROR 1146 (42S02): Table 'classwork.sp_cal' doesn't exist
D6_96715_bhagyesh>SELECT * FROM TEMPERATURE;
+------+------+
| F    | T    |
+------+------+
|   37 |  -16 |
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
D6_96715_bhagyesh>DELIMITER $$
D6_96715_bhagyesh>
D6_96715_bhagyesh>CREATE PROCEDURE SP_INCHES(IN TOTAL INT)
    -> BEGIN
    ->     DECLARE YARD INT;
    ->     DECLARE FEET INT;
    ->     DECLARE INCH INT;
    ->
    ->     SET YARD = TOTAL / 36;
    ->     SET FEET = (TOTAL % 36) / 12;
    ->     SET INCH = TOTAL % 12;
    ->
    ->     SELECT YARD AS YARD, FEET AS FEET, INCH AS INCHES;
    -> END $$
Query OK, 0 rows affected (0.06 sec)

D6_96715_bhagyesh>
D6_96715_bhagyesh>DELIMITER ;
D6_96715_bhagyesh>
D6_96715_bhagyesh>CALL SP_INCHES(124);
+------+------+--------+
| YARD | FEET | INCHES |
+------+------+--------+
|    3 |    1 |      4 |
+------+------+--------+
1 row in set (0.03 sec)
```

5. Create a procedure to read two real numbers and tell whether the

product of the two numbers is equal to or greater than 100.

```
D6_96715_bhagyesh>DELIMITER $$
D6_96715_bhagyesh>CREATE PROCEDURE sp_product_check(IN num1 DOUBLE, IN num2 DOUBL
E)
    -> BEGIN
    -> DECLARE product DOUBLE;
    -> SET product = num1 * num2;
    ->  IF product >= 100 THEN
    -> SELECT 'Product is equal to or greater than 100' AS Result, product;
    -> ELSE
    ->   SELECT 'Product is less than 100' AS Result, product;
    ->  END IF;
    -> END $$
Query OK, 0 rows affected (0.04 sec)

D6_96715_bhagyesh>DELIMITER ;
D6_96715_bhagyesh>CALL sp_product_check(10, 12);
+----------------------------------------+---------+
| Result                                 | product |
+----------------------------------------+---------+
| Product is equal to or greater than 100 |     120 |
+----------------------------------------+---------+
1 row in set (0.03 sec)

Query OK, 0 rows affected (0.04 sec)
```